TRY IT OUT: DETECT MULTIPLE OBJECTS IN A VIDEO FEED

Computer vision detects the position and type of objects within video frames, assigning bounding boxes and confidence scores. This allows digital twins to represent the location and status of physical objects like people, vehicles, or packages.

By running object detection at the edge, only the processed information (e.g., 'vehicle detected at entrance') is transmitted, not the full, high-bandwidth video stream.

You Only Look Once (YOLO - https://www.v7labs.com/blog/yolo-object-detection) is a popular, high-speed convolutional neural network model that predicts multiple object bounding boxes in a single pass. You can try the YOLO version 8 model out yourself with the code shown in listing 2.3, run on a computer equipped with a webcam. Whereas this code simply overlays the video feed with the detected objects, in a complete application, you would transmit the detected object events to your digital twin application (at a much lower cost than transmitting the full video feed).